

Deterministic geometric Savitzky-Golay dynamics estimator

Shared parameter update: reference -> estimated

mass [kg]	2.351597591 -> 2.557856401	delta=+0.206259	
ratio=1.0877101			
CoG x [m]	-0.002024708562 -> -0.03363213971	delta=-0.0316074	
ratio=-			
CoG y [m]	-3.052657894e-05 -> -0.04112920516	delta=-0.0410987	
ratio=-			
CoG z [m]	0.009509749599 -> 0.05558975391	delta=+0.04608	ratio=-
rotor effectiveness 1	1 -> 1.307535439	delta=+0.307535	ratio=1.3075354
rotor effectiveness 2	1 -> 0.7570064381	delta=-0.242994	ratio=0.75700644
rotor effectiveness 3	1 -> 0.6122215255	delta=-0.387778	ratio=0.61222153
rotor effectiveness 4	1 -> 1.047721063	delta=+0.0477211	ratio=1.0477211
rotor command lag [s]	0.004997730255 -> 0.1526245244	delta=+0.147627	
ratio=-			

Inertia matrix [kg m^2]

reference:

[0.06500006148, -7.27899253e-07, 1.901508003e-05]
[-7.27899253e-07, 0.06495265634, 5.9167305e-08]
[1.901508003e-05, 5.9167305e-08, 0.1289921107]

estimated:

[0.5698718645, 0.03618779132, 0.3743091704]
[0.03618779132, 0.8248187266, -0.05504606101]
[0.3743091704, -0.05504606101, 0.2656984664]

Principal moments: reference -> estimated

principal moment 1 [kg m^2]	0.06495264516 -> 0.008443187865	delta=-0.0565095	
ratio=0.1299899			
principal moment 2 [kg m^2]	0.06500006701 -> 0.8217513488	delta=+0.756751	
ratio=12.642315			
principal moment 3 [kg m^2]	0.1289921163 -> 0.8301945208	delta=+0.701202	
ratio=6.4360098			

Optimization

parameter-estimation prior: none (data-only objective)
joint dynamics loss: 835.733074432
selected strict-ZOH rotor lag [s]: 0.152624524376

Bag failure_1_single

reconstruction position RMSE [m]: 77.6031142674
reconstruction position component RMSE [m]: [9.212803 8.60222 76.572642]
reconstruction orientation RMSE [deg]: 142.755038528
gyro consistency RMSE xyz [rad/s]: [46.662538 20.72637 62.768614]
specific-force consistency RMSE xyz [m/s^2]: [140.623937 345.468651 114.615993]
inferred external force RMS xyz [N]: [10.427923 10.47115 8.696303]
inferred external torque RMS xyz [N m]: [50.683052 80.373022 29.486241]
bag dynamics loss: 835.733074432
gimbal command lag [s] 0.004998683929 -> 0.046265164

Recorded command timestamp intervals (data-derived)

Bag failure_1_single
rotor: median dt=0.00499773026 s (200.091 Hz), min=2.00271606e-05 s, max=0.0138893127 s
gimbal: median dt=0.00499868393 s (200.053 Hz), min=4.529953e-06 s, max=2.48282003 s

Selected optimizer diagnostics
status=0 success=False nfev=80 njev=72
message=The maximum number of function evaluations is exceeded.
reported optimality=0.6993044507030248
initial cost=835.7592113256753
final cost=835.7330744320466
cost reduction=0.026136893628745383
coordinate step L2=3.0791821753654385
final gradient inf-norm=0.6993044507030248
gradient tolerance satisfied=False
final Jacobian rank=13
final Jacobian cond(nonzero)=264480466.89701566
max directional finite-difference Jacobian relative error=3.2830777040459686e-10